KBAI Assignment:
Getting to Know Jill Watson

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***Abstract*—**(Optional) Replace this text with a brief summary of your findings from interacting with Jill Watson across all three parts of this assignment. Delete this entire abstract block if you choose not to include one.

1 CONCEPT EXPLORATIONS

1.1 Activity A: The four schools and grand challenges (Chapter 1)

Choose a modern AI application (e.g., a medical triage system, a content recommendation engine, a fraud detection system). Present it to Jill Watson and have a conversation that surfaces how JW applies the Four Schools framework and whether it grounds the Explanation Enigma in the specific terms of Chapter 1. Then, write 2–3 paragraphs addressing the report question: Does Jill Watson correctly apply the Four Schools categorization to a non-trivial example? Does it ground the Explanation Enigma in Chapter 1's specific terms, or drift toward generic AI discourse? Where does it succeed and where does it fall short?

1.2 Activity B: Problem-solving under constraint (Chapters 3 & 4)

Define your own Blocks World scenario with an initial state and a goal state where MEA is likely to encounter a local optimum. Present it to Jill Watson and have a conversation that surfaces two distinct things: (1) how MEA behaves at the local optimum and why its logic traps it there, and (2) how the Smart Generator in Generate & Test avoids that failure through pruning and what the balance of responsibility between Generator and Tester means in practice. Treat these as separate mechanisms. Then, write 2–3 paragraphs addressing the report question: Can Jill Watson articulate the balance of responsibility between the Generator and the Tester? Does it correctly diagnose the local optimum, and does its proposed pruning strategy follow the course definition of a Smart Generator?

1.3 Activity C: Frame-based understanding (Chapter 6)

Provide a short ambiguous scenario of your own design (e.g., "A visitor arrived at the building and was escorted to a room"). Present it to Jill Watson and have a conversation that makes visible the distinction between what you stated and what JW assumed using Default Values, and how Top-Down Processing governed those assumptions. A response that produces a Frame without explaining the source of each filler has not met the conceptual target. Then, write 2–3 paragraphs addressing the report question: Does Jill Watson distinguish between information stated in your scenario and information it assumed via defaults? Does it make that distinction transparent, or do you have to infer it? What does this reveal about the strengths and limitations of Frame-based representation for handling ambiguity?

2 COMPARISON ANALYSIS

2.1 Comparison matrix

Give Jill Watson the scene "A small blue cube is supported by two large red bricks, and a yellow sphere rests on top of the cube" with the prompt: "According to the KBAI ebook, construct a Semantic Network for this scene. Explicitly list the Lexicon (nodes) and the Structural Specifications (directed, labeled links) as defined in Chapter 2." Then give the same scene to a general LLM with the prompt: "Describe the spatial relationships between these objects in a structured way." Fill in both columns of the matrix below based on their responses.

2.2 Analysis

Write one paragraph addressing the report question: The two prompts differ intentionally — Jill Watson was anchored to the KBAI ebook; the general LLM was not. Does that prompting difference fully explain any quality gap you observed, or does the underlying architecture (RAG vs. general language model) appear to be doing independent work? If the general LLM produced a formally correct Semantic Network despite the weaker prompt, what does that imply? If Jill Watson underperformed despite the stronger prompt, what does that suggest about the limits of RAG-based grounding?

3 SOCRATIC AUDIT REFLECTION

Complete the 5-question Socratic audit with Jill Watson by pasting the instruction block from the comments above (or from the assignment page) into a fresh JW session. Then, write 2–3 paragraphs addressing the report question: If you were guided through a Socratic exchange, describe one instance where it corrected or deepened your understanding, and identify the specific question JW asked that shifted your thinking. If you answered all five correctly on the first attempt, reflect on which question you found most counterintuitive and what you would expect a skilled tutor to probe if you had answered incorrectly. In either case: if Jill Watson drifted from the question sequence or dropped its Socratic role entirely, describe when it happened and what that behavior suggests about the architectural difference between a stateless LLM and a human tutor who maintains conversational context.

4 REFERENCES

5 CHAT TRANSCRIPTS

IMPORTANT: This section is a required and graded deliverable. You must include the complete chat transcripts for every interaction across all three parts of this assignment. Incomplete or missing transcripts may result in point deductions. Six transcripts are required: three from Part 1 (JW only), two from Part 2 (one JW, one general LLM), and one from Part 3 (JW). This section does not count against the page limit.

5.1 Part 1 transcripts

5.1.1 *Activity A — Jill Watson*

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[Paste your Jill Watson chat transcript for Activity A here.]
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5.1.2 *Activity B — Jill Watson*

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[Paste your Jill Watson chat transcript for Activity B here.]
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5.1.3 *Activity C — Jill Watson*

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[Paste your Jill Watson chat transcript for Activity C here.]
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5.2 Part 2 transcripts

5.2.1 *Jill Watson*

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[Paste your Jill Watson chat transcript for Part 2 here.]
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5.2.2 *General LLM*

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[Paste your general LLM chat transcript for Part 2 here.]
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5.3 Part 3 transcript

5.3.1 *Socratic audit — Jill Watson*

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[Paste your Jill Watson Socratic audit transcript here.]
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6 APPENDICES (OPTIONAL — NOT GRADED)

IMPORTANT: This section is entirely optional. Nothing in the appendices will be considered for grading. All graded content must appear in the main body (Sections 1–3). The body text must be sufficient to answer every question on its own. Appendices are provided only for optional supplementary material (e.g., additional diagrams, extended analyses). They do not count against the page limit. If you have no appendices, delete this entire section.

JDF FORMATTING EXAMPLES (DELETE BEFORE SUBMISSION)

This section is provided as a reference for JDF formatting conventions. Delete it entirely before submitting your assignment.

Figure example

To include a figure, uncomment and adapt the code below:

To reference a figure in text: "As shown in Figure 1..." (use `\ref{fig:your_label}` once the figure is uncommented).

Figures should always be centered, referenced in text, and include a descriptive caption beneath them. If the figure has a white background, consider adding a $\frac{1}{4}$-point black border for legibility.

Table example

Table captions go *above* the table ("figures at the foot, tables at the top").

In-line text formatting

Bold and *italics* for emphasis. Hyperlinks: link text. In-line code: `code here`. Superscripts $^{\text{like this}}$ and subscripts $_{\text{like this}}$.

Quotes

In-line (fewer than 3 lines): "Heavy use of peer grading would compromise [the school's] reputation" (Joyner, 2016).

Block quote (3+ lines):

> "Whether or not the grades generated by peers are reliably similar to grades generated by experts is only one factor worth considering, however. Student perception is also an important factor."

Lists

Bullet points:

- First bullet point item
- Second bullet point item

Numbered list:

1. First numbered item
2. Second numbered item

Citations

Parenthetical: `\citep{key}`. In-text: `\citeauthor{key} (\citeyear{key})`. Multiple: `(\cite{key1}; \cite{key2})`. Place citations close to the claim, not just at paragraph end.

Footnotes

In-line citations are preferred over footnotes. If you do use footnotes, they should be 8.5 pt with 14 pt line spacing. Example: `\footnote{Your footnote text here.}`

Reference list

References are handled by `\printbibliography`. They should be numbered, alphabetical by first author's last name. Same author with multiple papers: chronological order (oldest first). Same author and year: append a letter (e.g., 2018a, 2018b). Only include works cited in-line. The reference list does not count against the page limit.

Heading 4 example

Run-in sidehead—*Heading 4* is a run-in sidehead in bold italics, followed by an em dash, flowing into the paragraph text. Use it as a list style rather than a heading style.

Table 1—Comparison matrix: Jill Watson vs. General LLM.

Criterion	Jill Watson	General LLM
Groundedness: Does the response use course-specific terminology (Lexicon, Structural Specifications, directed, labeled links), or fall back on generic AI/graph language?	Fill in your observation.	Fill in your observation.
Representational Rigor: Does the response correctly separate nodes (Lexicon) from directed, labeled links (Structural Specifications), or conflate them into a narrative or flat list?	Fill in your observation.	Fill in your observation.
Reasoning Transparency: Does the response explain why links are directed and what relationship each link encodes, or simply list connections without justification?	Fill in your observation.	Fill in your observation.

Table 2—Example table with mathematical constants.

Name	Symbol	Approximation	Description
Golden ratio	ϕ	1.618	Ratio where ϕ to the whole equals the reciprocal to 1
Euler's number	e	2.71828	Exponential growth constant
Archimedes' constant	π	3.14	Ratio of circumference to diameter